

Numerical Optimisation Project

Part A: Basin Planning Model

This report presents the Part A basin planning model. It briefly covers the approach used to develop the model, including model elements and how to run the model. It also includes a brief outline of how the model was verified.

Focus of this report

Clear formulation, clean implementation blocks, and result graphs that can be discussed quickly during an oral exam.

Contents

1	Part A: Farm Planning Model	2
1.1	Executive Summary	2
1.2	Model in One View	2
1.3	Input Data	2
1.4	Decision Variables	3
1.5	Model Logic	3
1.6	Graphs and Results	3
1.7	Implementation Notes	4
1.8	Interpretation	4
1.9	Current Scope	4

Part A: Farm Planning Model

Executive Summary

Part A solves a crop planning problem with irrigation and labour decisions. The model chooses the crop mix, pumping, hiring, and sales strategy that maximizes annual gross margin under the farm constraints.

Objective value	Pump licence	Final reservoir	Peak labour
\$122,077.77	100%	10,000 m ³	125 h

Model in One View

The problem uses three sets: land blocks B , crops C , and periods T . The model selects area variables $x_{b,c}$, pumping p_t , reservoir level r_t , hired labour h_t , and sales variables for each crop.

- Objective: maximize revenue minus crop cost, hired labour cost, and pumping cost.
- Key trade-off: more irrigation and labour allow higher-value crops, but each resource is capped.
- Structure: the code is split into clear blocks in `src/model.jl` so each component can be modified independently.

Input Data

The implementation reads the following CSV files:

- `blocks.csv`: land blocks, area, and soil class.
- `crops.csv`: crop parameters such as variable costs and farm share limits.
- `yields.csv`: yield by crop and soil class.
- `calendar.csv`: irrigation and labour requirements by period.
- `labour.csv`: family labour, hired labour capacity, and hired labour cost.
- `water.csv`: inflows and pumping limits.
- `markets.csv`: prices and contract information.
- `parameters.csv`: global constants such as reservoir capacity and pumping cost.

These inputs are loaded into a single `FarmData` structure in `src/data.jl`.

Decision Variables

$x_{b,c} \geq 0$	area of crop c on block b ,
$p_t \geq 0$	water pumped in period t ,
$r_t \geq 0$	reservoir level at the end of period t ,
$s_t \geq 0$	spill in period t ,
$h_t \geq 0$	hired labour in period t ,
$q_c^{\text{contract}} \geq 0$	contracted sales of crop c ,
$q_c^{\text{spot}} \geq 0$	spot sales of crop c .

Model Logic

Land allocation Each block is fully assigned across crops, and each crop has a maximum farm share.

Production and sales Production is computed from area and yield. It is then split into contract and spot sales when a market exists.

Water system The reservoir evolves period by period using inflow, pumping, irrigation use, and spill. Reservoir capacity and annual pumping licence both bind the water system.

Labour system Family labour and hired labour together cover crop labour needs, while hired labour itself is capped each period.

Graphs and Results

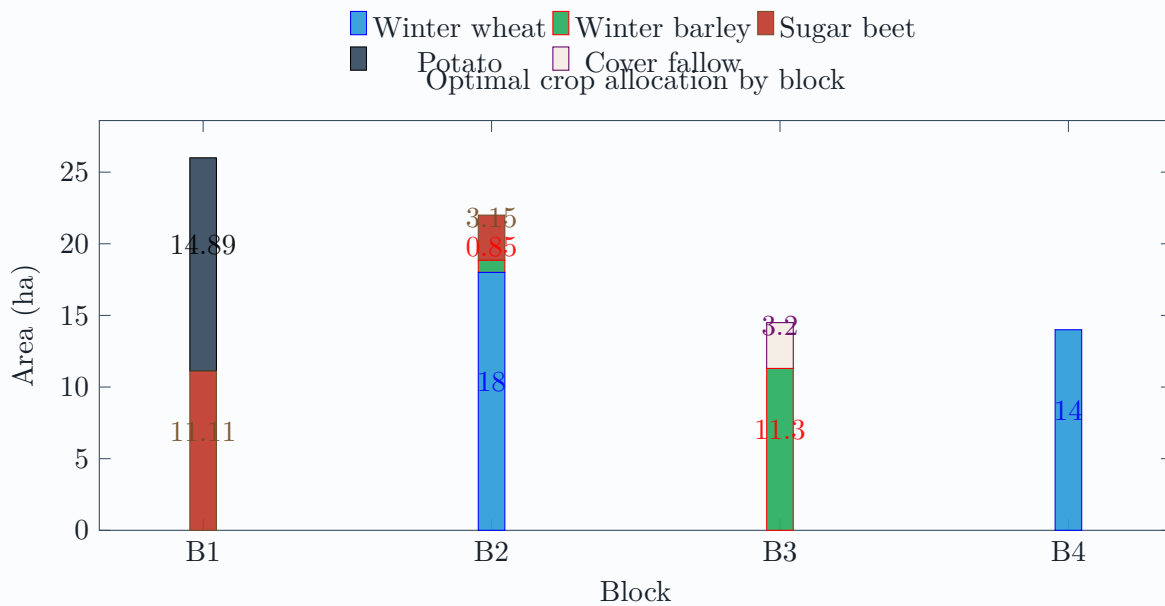


Figure 1: The optimal crop mix concentrates value in sugar beet and potato on block B1, uses winter cereals on B2 and B4, and leaves a cover-fallow area on B3.

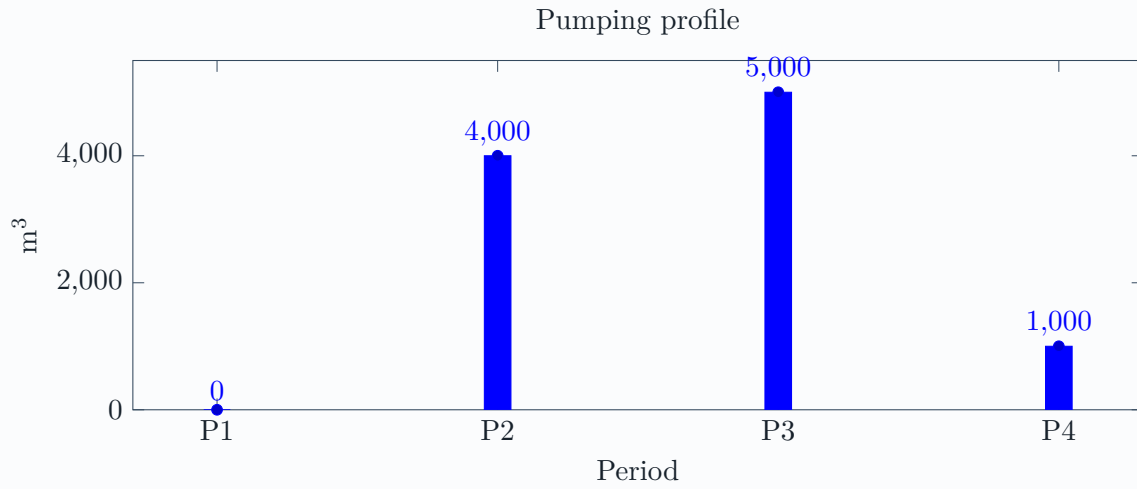


Figure 2: Water use is concentrated in P2–P4, and the annual pumping licence is fully used.

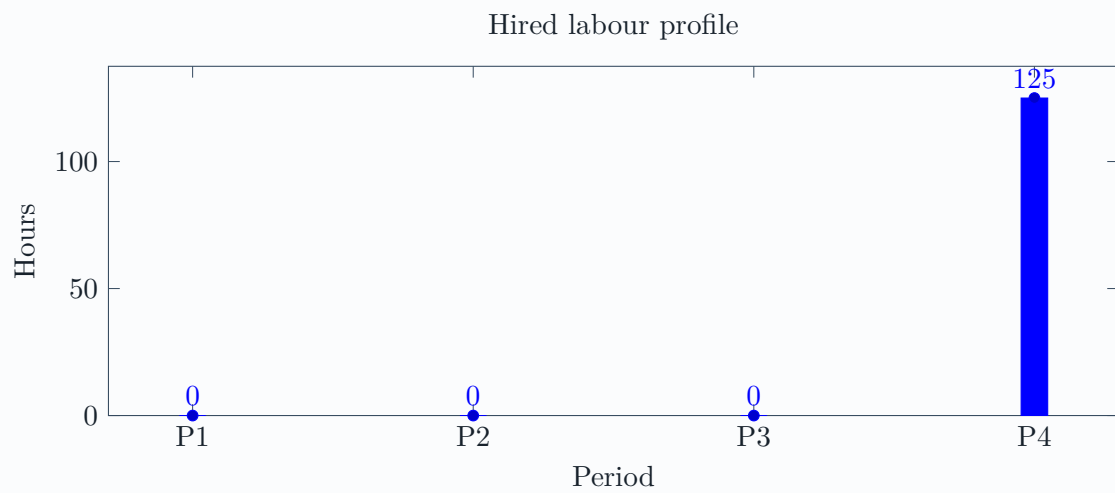


Figure 3: Hired labour is only required in the final period, where the labour cap binds.

Table 1: Selected solution values from the optimal Part A run

Crop	Contracted sales	Spot sales	Total	Comment
Winter wheat	302.40	0.00	302.40	Contracted
Winter barley	98.72	0.00	98.72	Contracted
Grain maize	42.04	0.00	42.04	Contracted
Sugar beet	1199.30	0.00	1199.30	Contracted
Potato	700.00	0.00	700.00	Contracted
Field beans	0.00	0.00	0.00	Not used

Implementation Notes

The code is organized in `src/model.jl` into separate blocks for data preparation, land constraints, market constraints, resource constraints, and the objective function.

That structure is intentionally kept simple so it can be explained and modified quickly during an oral exam.

Interpretation

The optimal plan uses the available water licence fully, hits the reservoir terminal target, and requires hired labour only in the last period. The crop pattern shows the model balancing high-margin crops against water and labour scarcity.

Current Scope

This report intentionally covers only Part A for now. The document and codebase can be extended later with sensitivity analysis, Part B/C formulations, or comparison against alternative crop plans.